

Problem Set 5: Web Data Aggregation

Econ 5253 — Spring 2026

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1 Task 1: Web Scraping — BLS Mass Layoffs Statistics

Data Source

The data were scraped from the Bureau of Labor Statistics (BLS) Extended Mass Layoffs Statistics release page:

<https://www.bls.gov/news.release/mslo.t01.htm>

This page presents a summary table of mass layoff events, the number of initial unemployment insurance (UI) claimants, and related industry breakdowns. The table is published as static HTML with no public API endpoint, making it a natural target for BeautifulSoup-based scraping.

Scraping Procedure

The script `PS5_Zhou.py` uses Python's `requests` library to fetch the page and `BeautifulSoup` (from the `bs4` package) to parse the HTML. The relevant table is identified by its CSS class "`regular`", which BLS uses consistently across its statistical release pages. Each `<tr>` element is iterated to extract cell text, and the result is assembled into a `pandas` `DataFrame` and exported as a CSV.

I did not rely on external tutorials beyond the official BeautifulSoup documentation (<https://www.crummy.com/software/BeautifulSoup/bs4/doc/>) and the BLS website's own HTML source inspection.

Why This Data Is Relevant

My current research focuses on labor economics topics in accounting, particularly the stock market and accounting consequences of mass layoff announcements. The BLS Extended Mass Layoffs program directly measures the events I study: employer-initiated separations of 50 or more workers from a single establishment over a five-week period. Having a clean, programmatically-updated version of this table is directly useful for:

- Benchmarking the time-series distribution of mass layoff events against the RavenPack news-based layoff sample I use in my event study.
- Identifying macroeconomic periods of elevated layoff activity (e.g., the 2001 recession, the 2008–2009 financial crisis, and the 2020 COVID shock) as controls or subperiod filters in panel regressions.
- Providing descriptive context for the introduction and motivation sections of my seminar paper.

2 Task 2: BLS Public Data API v2

API and Series

The BLS provides a free, unauthenticated REST API (v2) at

<https://api.bls.gov/publicAPI/v2/timeseries/data/>

I queried two series from 2000 through 2024:

1. **LNS14000000** — Unemployment Rate, seasonally adjusted (Current Population Survey).
2. **JTS0000000000000000LDR** — Layoffs and Discharges Rate, Total Nonfarm, seasonally adjusted (Job Openings and Labor Turnover Survey, JOLTS).

No registration key is required for the public API tier. Requests are sent as JSON `POST` payloads. The `calculations` flag was set to `true` to retrieve month-over-month changes alongside levels.

Selected Findings

Table 1 reports annual averages of both series for selected years.

Several patterns stand out. First, the layoffs and discharges rate spiked most sharply in April 2020 (reaching approximately 7.6%), reflecting the sudden, pandemic-induced employment separation wave—a period qualitatively different from the more gradual recessionary layoff cycles of 2001–2002 and 2008–2009. Second, despite a historically low unemployment rate in 2022–2023 (around 3.6%), the layoffs rate remained modest, consistent with tight labor market conditions limiting employer separations. Third, the positive unconditional correlation between the two series over the full sample (the unemployment rate tends to rise following periods of elevated layoffs) aligns with the labor market dynamics documented in Blanchard and Diamond (1990) and motivates the use of macro controls in layoff-announcement event studies.

Packages Used

- `requests` — HTTP requests for both scraping and API calls.

Table 1: Annual Average Unemployment Rate and Layoffs & Discharges Rate, Selected Years (2000–2024)

Year	Unemployment Rate (%)	Layoffs & Discharges Rate (%)
2000	4.0	1.3
2001	4.7	1.6
2002	5.8	1.5
2007	4.6	1.3
2008	5.8	1.5
2009	9.3	1.8
2010	9.6	1.5
2019	3.7	1.2
2020	8.1	2.2
2021	5.4	1.0
2022	3.6	1.0
2023	3.6	1.1
2024	4.0	1.1

- `beautifulsoup4` (`bs4`) — HTML parsing for Task 1.
- `pandas` — Data manipulation and CSV export.
- `json` — Serialization of API request payload.